Credit Risk Report

Introduction

Credit risk modelling plays a central role in financial decision-making, where lenders must assess the likelihood that a borrower will default on a loan. Accurate classification of borrowers into higher and lower risk categories is essential for managing losses, pricing loans appropriately, and maintaining portfolio stability.

This project develops classification models to predict the probability of default using the German Credit Dataset. The analysis explores borrower characteristics, constructs classification models, and evaluates their performance in differentiating between high and low risk borrowers.

The analysis also identifies key factors of default risk and considers how these insights could support real-world lending decisions.

Preparation

Data Preparation & Preprocessing

Before modelling, the dataset must be prepared to ensure it is suitable for statistical analysis. The dataset contains both numerical and categorical variables describing borrower characteristics. As most statistical models require numerical inputs, categorical variables must be encoded appropriately, and the data must be structured for supervised classification.

The German Credit Dataset consists of a mixture of numerical and categorical variables describing borrower characteristics, such as credit history, loan purpose and financial status.

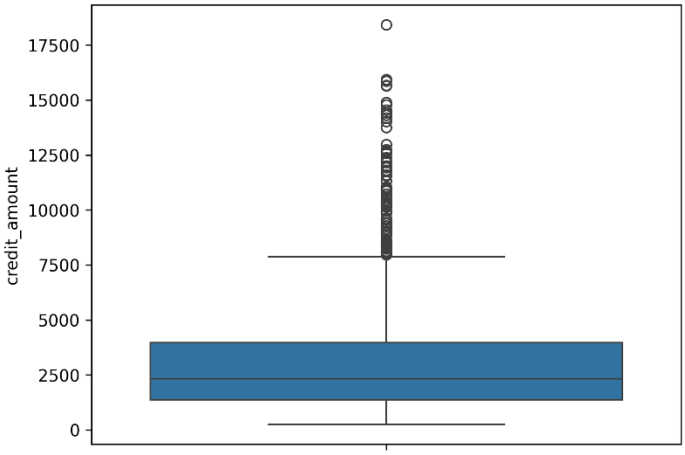
As the dataset does not include column labels, they are added manually to allow for easier interpretability and support analysis.

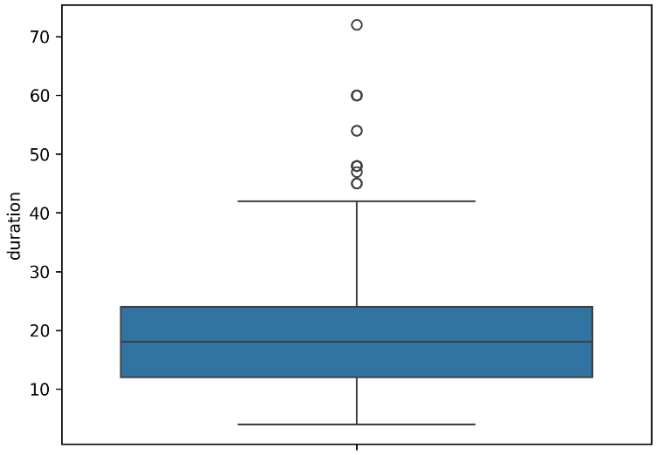
Data Quality

Using pandas, the data type of each column can be checked, while also identifying any null entries. This is important to check as many models will break if any data is missing.

There were no missing values identified across the dataset, but other common errors in datasets are duplicated entries, inconsistent formatting (such as data formatting), and inaccurate data, so these were all checked for in Python.

No duplicate observations or anomalies were found, however, some potential outliers were found in the variables ‘credit_amount’ and ‘duration’, so their box plots were investigated:





Despite the box plots marking multiple points as outliers for both variables, this does not mean that these points are errors in the data. Thinking about each of the variables, both can have very large values, as large 'credit_amount' values would correspond to very large loans, and it isn't unheard of to have very long loan periods for a loan's 'duration'.

This means that it is likely that these are real observations, and not errors, so they should probably be kept.

However, by looking at the box plots, especially the 'credit_amount' boxplot, we see that the data looks positively skewed, with a concentration of smaller values and a long right tail.

In cases where there is lots of right-skew, a log transformation can be applied to the 'credit_amount' values, which can often improve a model’s performance. This will be explored more later.

The Target Variable

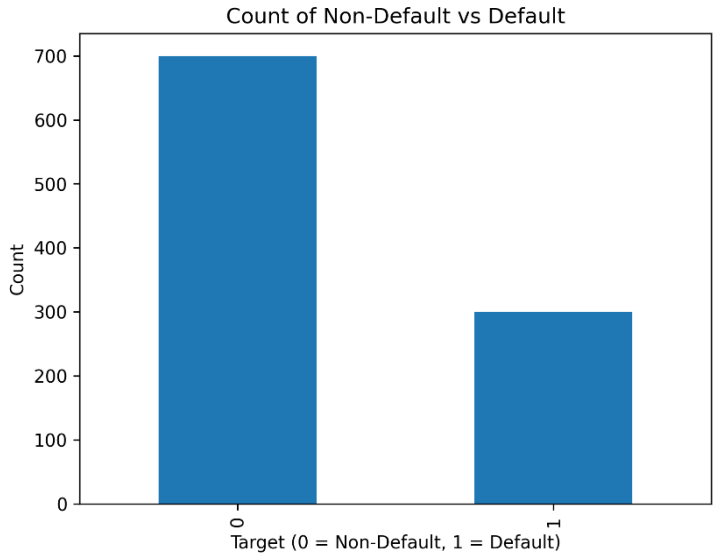
By examining the head of the dataset, it can be seen that the 'target' variable takes values between 1 and 2.

In the context of this data, a value of 1 indicates the borrower is a 'good' borrower, and so is unlikely to default, while a value of 2 indicates that the borrower is a 'bad' borrower and is likely to default.

Regression models predict values of 0 and 1, or -1 and 1, so to fit this requirement the values are changed so that 0 means non-default and 1 means default.

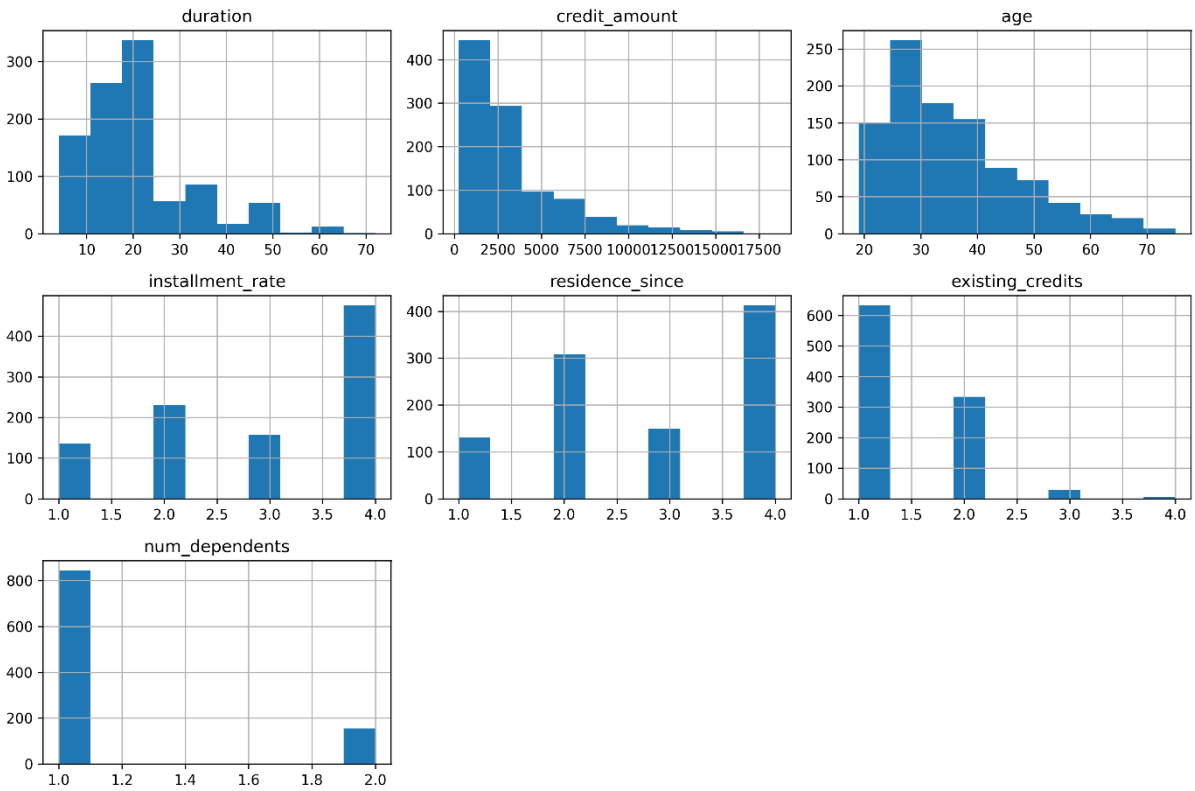
Exploratory Analysis

The analysis begins by examining at the distribution of the target variable and some key numerical features to understand the data better.



This reflects real-life, where most borrowers do not default. However, this distribution of data may lead to a poor model, as it has less information on what makes borrowers likely to default.

Some histograms of the other variables can be examined:



These histograms further confirm that some of the variables may be skewed.

Next, we can see what the average for some of the variables are, when categorised by their target value - so categorised by whether they are a good borrower or not.

target	duration	credit_amount	age
0	19.21	2985.46	36.22
1	24.86	3938.13	33.96

The borrowers who do not default are on average a higher age than those who do, which is expected as with age typically comes financial security, and better career progression.

The data says that on average the borrowers who defaulted had the loans over a longer duration. This is expected, as those who can pay back loans would likely go for a shorter loan duration, whereas those less confident in paying it back may want more time.

Finally, the average credit_amount variable is what we would expect to see. Borrowers who defaulted had, on average, higher credit amounts. This is consistent with expected financial behaviour.

Data Preprocessing

Before building the model, categorical variables need to be converted into numerical form. This ensures that all variables are in a numerical format suitable for statistical modelling.

As noted before, a lot of the data values are preceded by the letter A, which means regression models would not understand them numerically. Therefore, we change them to simple numbered categories, as can be seen in the table below. One hot-coding is used to convert categorical variables into numerical form. Even after one hot-coding, some variables remain in Boolean form, however, Boolean variables are automatically encoded as binary values, ensuring compatibility with statistical models.

In a supervised learning setting, the data is separated into input variables and a target variable. Hence, the dataset is split the into inputs (X), where one input would be a row of all the variables, excluding the 'target' variable, and then an output (y), which is then that row's corresponding 'target' variable.

The model is trained to learn the relationship between borrower traits and the probability of default risk.

To measure the effectiveness of our model on unseen data (i.e. any data that was not used to train it), the data is split into training and testing data. The model is produced using only the training data, then it is evaluated on unseen data using the test set.

Following preprocessing, the dataset is ready for model development, beginning with logistic regression, which is a baseline classifier in risk.

Logistic Regression Model

Fitting the Model

The first model that will be used is the logistic regression model, a widely used and interpretable model in credit risk analysis.

Logistic regression works better when the variables are of a similar scale, so feature scaling is applied to the data, before the model is trained. Logistic regression doesn't have a direct 'solution' that can be found and is definite like in simple regression. Instead, it uses optimisation algorithms to improve the fit of the model by updating over lots of iterations.

We can explore the quality of the results when the maximum number of iterations change, as it may results in over or under training of the model, however 100 iterations seem sensible for now.

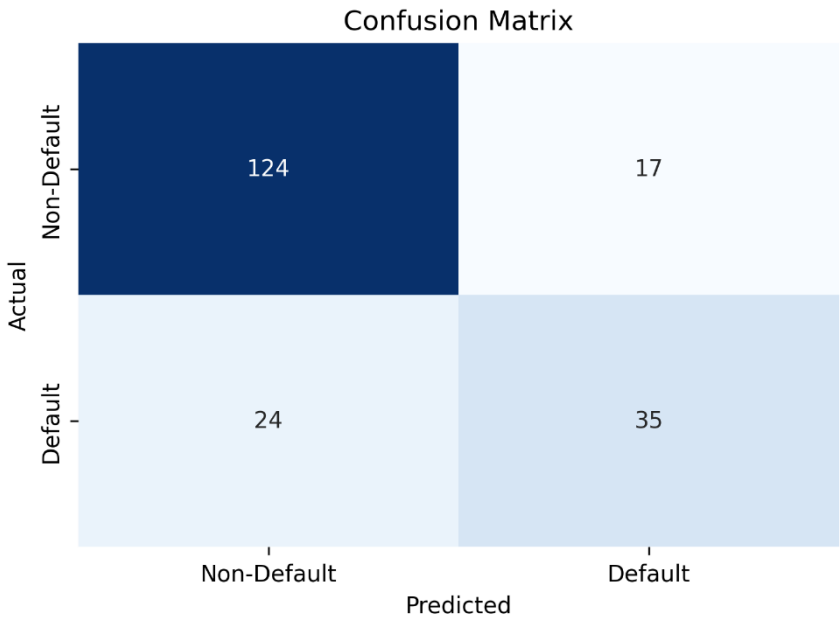
In the end it only took 24 iterations to find an optimal model.

Performance

To assess the model's performance, it will be made to produce the probabilities of defaulting, and the subsequent prediction of whether the unseen borrowers will default or not for the unseen test data.

Whether the unseen borrowers defaulted or not, is known, so the overall accuracy of the model's predictions can be found. The logistic regression model had a prediction success rate of 79.5%. This provides a useful baseline for accuracy ratings. However, false negatives (incorrectly classifying defaulters) are significantly more costly than false positives (incorrectly classifying non-defaulters), so it is important to examine where the model made its mistakes.

To see where the model made its mistakes, a confusion matrix can be plotted.



The test data consisted of 200 borrowers, of which 141 did not default their loans, while the remaining 59 did.

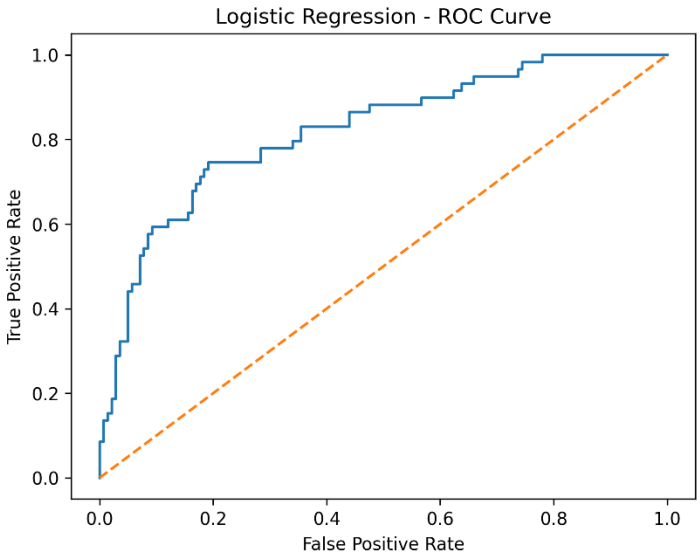
Out of the 141 non-defaulters, the model correctly predicted 124 would not default and incorrectly predicted that 17 would default. That means it correctly identified non-defaulters 87% of the time. Similarly, it correctly identified defaulters 59% of the time. The model performs significantly better at identifying non-defaulters than defaulters, reflecting the class imbalance in the data. This is known as the recall for each class.

Another statistic to examine is how accurate was the model when it predicted default or non-default. Looking at the confusion matrix, it can be calculated that given the model predicted default on a borrower, it was correct 67% of the time. Similarly, given the model predicted non-default for a borrower, it was correct 84% of the time. This is known as precision.

The difference in these statistics is subtle but is important to differentiate between the two. Recall measures how well the model identifies actual outcomes, while precision reflects how reliable the model's predictions are.

Both sets of statistics indicate that the model is more reliable when predicting non-default than default. In a risk context, this is particularly important, as failing to identify defaulters can lead to significant financial losses. A way to mitigate the model's inaccuracy will be explored later by adjusting decision thresholds to balance risk.

A curve that illustrates the trade-off between the true positive rate (recall) and the false positive rate across different classification thresholds is the ROC curve. The closer the curve is to the top left corner, the better the model is at distinguishing between defaulters and non-defaulters.



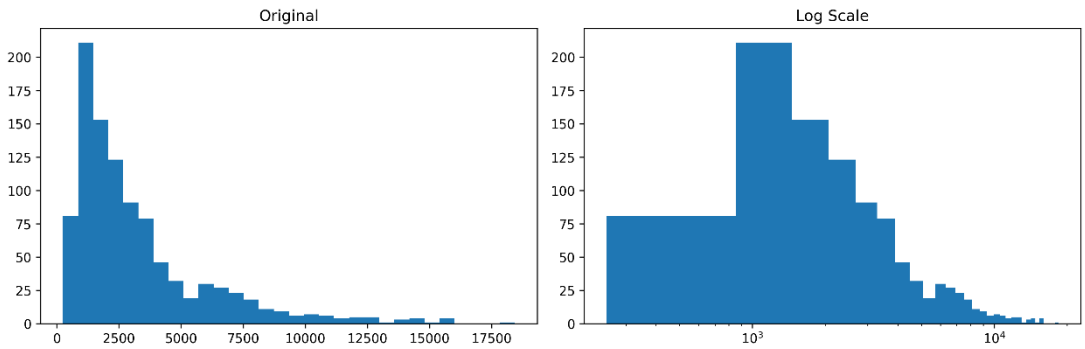
The ROC AUC (area under curve) gives a good indication of overall discriminatory performance. The logistic regression model achieves an AUC of approximately 0.82, which indicates a good discriminatory performance. This means that, given a randomly selected defaulter and non-defaulter, the model will correctly assign a higher probability of default to the defaulter around 82% of the time.

This suggests that the model is not perfect, however, it is effective at ranking borrowers by risk, which is valuable in credit risk applications where decisions are often based on thresholds rather than fixed classifications.

Skewed Data

As discussed earlier, certain variables seemed to have data that was skewed. This means that there were a greater number of small values and fewer very large values in the data, or vice versa. The histogram and boxplot of the 'credit_amount' variable highlights this well.

To make the data less skewed, a transformation can be applied to the skewed variables to make them more symmetric. As 'credit_amount' has lots of small values, but fewer large values, a log transformation seems appropriate. Looking at a histogram of the 'credit_amount' data on a regular scale, and a log scale should show the rough effect the transformation would have.



The data isn't perfectly symmetrical, but it is certainly less skewed than before. However, before the variable is scaled it is important to discuss the benefits and drawbacks of transforming data.

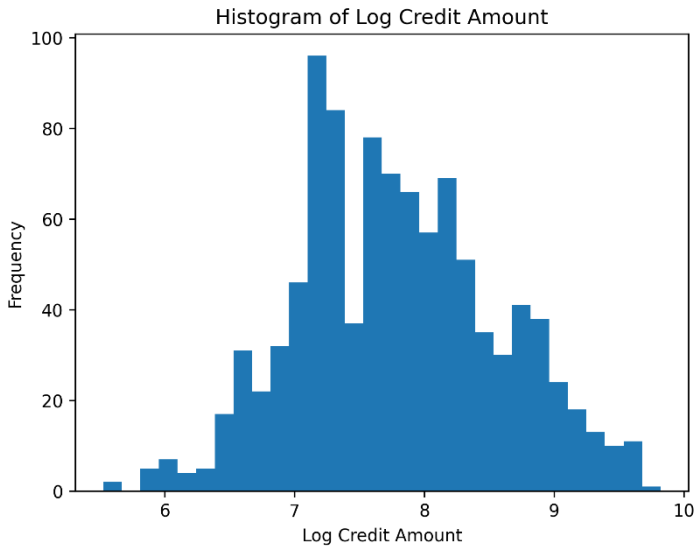
Discussing Transformations

Whether a variable should be transformed or not depends on how well it improves your model. Transforming variables doesn't always improve a model's performance, as transformations may cause some key patterns in the data to be lost, so they shouldn't be applied for the sake of it.

However, even if it does improve a model's performance it comes at the drawback of making interpretations of that variable's role in the model harder. Without a transformation it is easy to quantify how much a unit change in a variable affects the predicted probability, but when transformed the effect of the variable can get muddled.

Some models assume that the data isn't skewed (assumes the data is normalised more precisely), however logistic regression is not one of those models, so transforming the data isn't required to make the model work properly. However, if the data is very skewed it may affect the model's performance.

The only way to see how this will affect the model is to try. Whether the transformed variable is used over the regular variable can be determined afterwards.



The variable is now significantly more symmetrically distributed and is much less skewed, but to see if it was effective, the model needs to be fitted again on the dataset with the transformed variable.

After fitting the model, it is clear that the transformation did not change the model's performance. This means the transformation does not provide enough benefit to justify its inclusion, as it would only make interpretation harder later.

This suggests that the logistic regression was robust to the original skewed distribution and was a suitable choice for a first model to try.

Coefficient Analysis

Despite having used the logistic model twice, the structure of the model itself has not been explained. The logistic regression model is expressed as:

$$\log\left(\frac{\mathrm{p}}{1\text{-}\mathrm{p}}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Where p is the probability that a borrower defaults, and $x_1, x_2, \ldots, x_n$ are the n variables in the data. Then the values of $\beta_1, \beta_2, \ldots, \beta_n$ are known as the coefficients of their respective variables.

The coefficients for all the variables show how much the log-odds of default change given a unit change in that variable. If the coefficient is positive, then increasing that variable increases default risk. If it is negative, then increasing that variable decreases default risk. The larger the absolute value of the coefficient, the bigger the change in default risk when that variable changes.

Looking at the coefficients can tell us what the logistic regression has deemed the biggest factors affecting the likelihood of defaulting.

Below are the 10 variables with the largest positive coefficients, and the 10 variables with the largest negative coefficients from the fitted model:

Feature	Coefficient
Credit Amount	0.368
Installment Rate	0.349
Job A173	0.341
Job A174	0.257
Duration	0.247
Job A172	0.222
Existing Credits	0.200
Property A124	0.195
Property A122	0.143
Property A123	0.112

The variable with the largest positive coefficient is 'credit_amount'. This indicates that, holding other variables constant, higher loan amounts are associated with an increased

log-odds of default. This is intuitive, as higher credit amounts represent a greater financial burden on the borrower, increasing the likelihood of repayment difficulties.

The next biggest positive coefficient belongs to the variable 'installment_rate'. A higher instalment rate implies larger repayments relative to income, which could place more financial strain on the borrower and increase the risk of default.

The variables 'job_A173' and 'job_A174' have the next largest positive coefficients. These variables represent different job categories relative to a baseline group. Specifically, 'job_A173' corresponds to skilled employees, while 'job_A174' corresponds to highly skilled or management roles. The large positive coefficients indicate that, compared to the reference category, borrowers in these groups are associated with higher default risk. This does not inherently imply that these occupations lead to higher risk but may instead reflect underlying factors not explicitly captured by the model, such as differences in income stability, employment conditions, or financial behaviour.

After examining the variables that increase default risk, the variables that reduce default risk should be considered.

Below are the 10 variables with the largest negative coefficients from the fitted model:

Feature	Coefficient
Checking Account A14	-0.778
Credit History A34	-0.644
Purpose A41	-0.495
Personal Status Sex A93	-0.454
Purpose A43	-0.416
Purpose A42	-0.353
Age	-0.319
Other Installment Plans A143	-0.287
Credit History A32	-0.282
Credit History A33	-0.281

These variables likely capture aspects of financial stability, suggesting that borrowers with these characteristics are better able to meet their repayment obligations.

The largest negative coefficient corresponds to 'checking_account_A14', which indicates that the borrower has no checking account. Therefore, the model suggests that, on average and relative to a reference group, borrowers with no checking account are associated with a lower risk of defaulting. This may seem counterintuitive at first, but borrowers without a checking account may rely less on credit or be more

conservative borrowers. Alternatively, this variable may be capturing underlying factors not directly seen in the data.

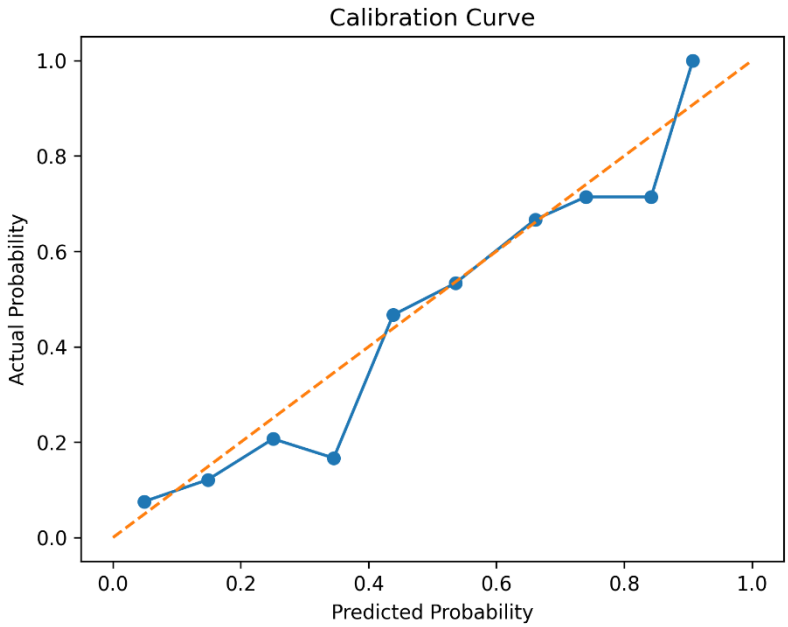
The variable 'credit_history_A34' has the next largest negative coefficient. This category represents borrowers with no previous credit issues, either having no prior credit or having repaid all previous loans in full. A large negative coefficient indicates that such borrowers are significantly less likely to default, which is expected, as past repayment history is a strong indicator of whether borrowers are worthy of more credit.

The variable 'purpose_A41', corresponds to loans that are for used cars, and is associated with lower default risk. This may reflect the relatively small size and practical nature of such loans compared to other borrowing purposes.

Probability Calibration

The model provides predicted probabilities of default, but in a risk context, well calibrated probabilities are crucial, as they are used to inform decision making, such as setting approval thresholds and estimating losses. It is therefore important to assess how well these probabilities reflect actual outcomes.

This can be assessed using a calibration curve.



The dashed line represents perfect calibration, where predicted probabilities match observed outcomes. If the model is well calibrated, the curve should be close to the dashed line. Deviations indicate that the model may over or under estimate default probabilities.

Some deviations from perfect calibration are seen, which tells us the predicted probabilities are not fully aligned with actual outcomes. However, this is to be expected as no model is perfect, and the deviations are moderate, suggesting reasonable calibration.

The biggest deviations on the graph are when the curve sits below the dashed line. This is when predicted probability is greater than the actual probability of default. This implies that the model is overestimating risk in those regions. This tells us that the model is slightly too conservative and may consider some borrowers to be riskier than they truly are.

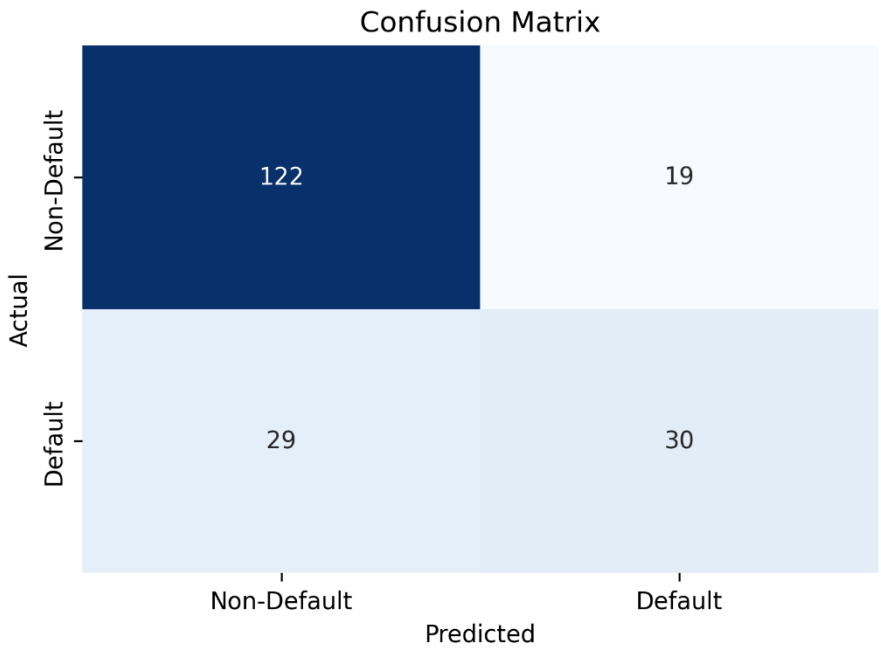
Decision Tree Model

Fitting and Assessing the Model

To compare the logistic regression model with a more flexible approach, a decision tree classifier was also fitted. Unlike logistic regression, a decision tree can capture non-linear relationships and interactions between variables, although this may come at the cost of interpretability and a higher risk of overfitting.

After fitting the model, the same performance measures used for the logistic regression model are looked at. The decision tree achieved an accuracy of 0.76 on the test data, which is lower than the logistic regression model's accuracy of 0.795. While accuracy provides a useful baseline, it is important to see how the model performs across different metrics.

The confusion matrix for the decision tree model is presented below.

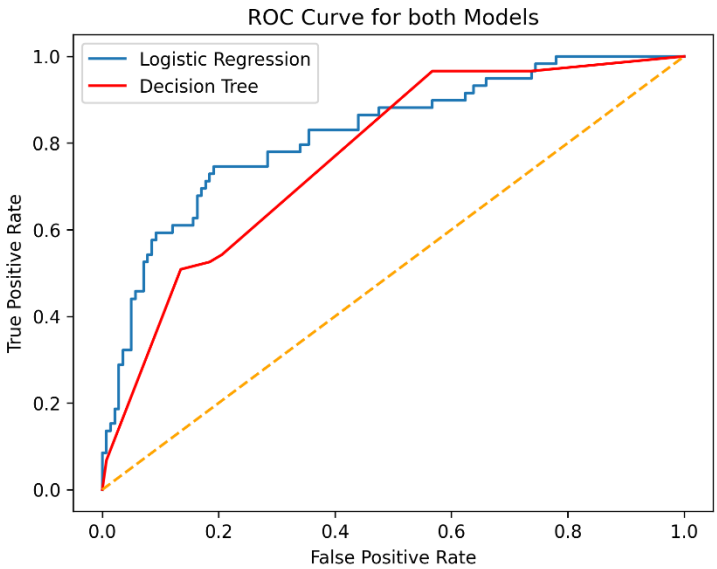


The decision tree model correctly identified 122 non-defaulters and 30 defaulters, while incorrectly classifying 19 non-defaulters and 29 defaulters.

This indicates that the model performs well at identifying non-defaulters, but struggles more with identifying defaulters, correctly identifying only 51% of them. This is lower than the logistic regression model, which had a recall of 59% for defaulters.

In terms of precision, when the decision tree predicts default, it is correct 61% of the time, which is also lower than the logistic model. Overall, this suggests that the decision tree model is less reliable for identifying high risk borrowers.

Comparing ROC curves for the two models also shows a weaker performance from the decision tree, as it is closer to the dashed line in the centre.

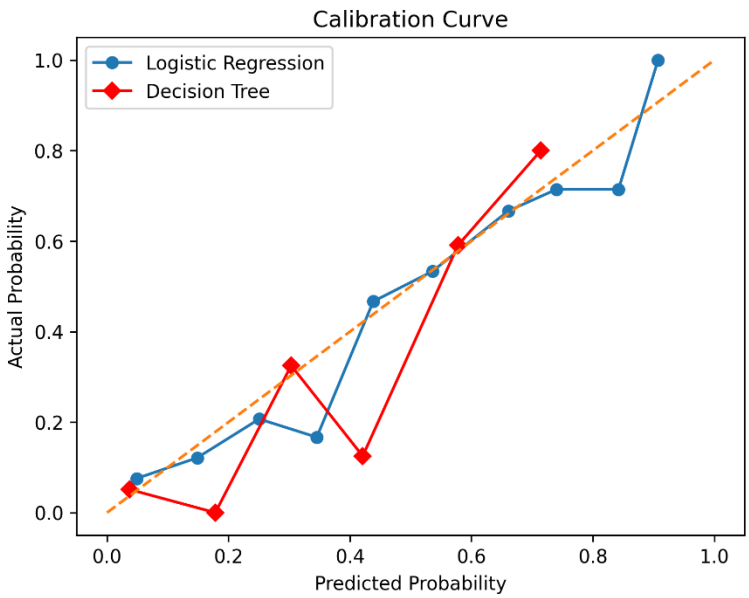


This can be measured objectively by comparing the ROC AUC of both models. The decision tree scored an ROC AUC of around 0.77. This indicates that the decision tree model is less effective at distinguishing between defaulters and non-defaulters, compared to the logistic model's AUC of 0.82. As many credit risk decisions rely on ranking borrowers rather than strict classification, this makes the decision tree less suitable in such risk environments.

Probability Calibration

The calibration curve for the decision tree model shows significant deviations from the diagonal line representing perfect calibration. The curve is also less smooth than the logistic regression's model, reflecting greater inconsistency in its probability estimates.

This indicates that the decision tree's predicted probabilities are less suited for decision-making in a risk environment, where well calibrated probabilities are essential. This puts the decision tree at a key disadvantage to the logistic regression.



Conclusions on the Models

The metrics demonstrate that the logistic regression model is superior to the decision tree, as it outperforms the decision tree across all metrics.

Additionally, due to the logistic regression's clear and interpretable formula, it allows direct insight into how each variable affects default risk. This is very important in a credit risk context, where understanding the drivers of default is essential for decision making.

While decision trees can also be interpreted by their structure, their predictions are based on a series of splits, which makes quantifying the effect of individual variables harder.

Overall, the logistic regression model performs more strongly across all the key evaluation metrics, and although the decision tree offers greater flexibility and the ability to model non-linear relationships, this did not translate to an improved performance. Given its stronger predictive performance, better calibrated probabilities, and greater interpretability, logistic regression is better suited for this credit risk project.

Threshold Analysis

Thresholds

The way that the models work, is that given a row of data (i.e. the profile of specific information on one borrower), they will predict a probability that the borrower defaults. Here lower probabilities indicate lower risk, and higher values indicate higher risk. The values will be between 0 and 1 as they are probabilities.

By default, a model will predict that a borrower will default if it predicts a probability of higher than 0.5. Here 0.5 is known as the probability threshold for the model, however, the threshold of 0.5 is completely arbitrary in this context. Can a better threshold be chosen?

In real life having a borrower default signifies a significant loss, whereas rejecting a good customer isn't as costly, so it is better to have a smaller margin of error with approving defaulters, if the trade-off is a larger margin of error with approving better borrowers. However, it is necessary to determine a threshold that balances this trade-off. It is hardly a realistic business model to reject every customer to be certain none ever default. Therefore, some alternative thresholds should be explored.

First, some confusion matrices for each threshold can be examined to see what occurs at various threshold values.

Threshold	TN	FP	FN	TP	Approvals	Approval Rate	Default Rate Among Approvals
0.1	49	92	4	55	53	26.5%	7.5%
0.2	78	63	8	51	86	43.0%	9.3%
0.3	101	40	14	45	115	57.5%	12.2%
0.4	116	25	17	42	133	66.5%	12.8%
0.5	124	17	24	35	148	74.0%	16.2%

The model is predicting defaults, so true positives (TP) are the defaulters that were successfully predicted. Similarly true negatives (TN) are the non-defaulters that were identified as such. The danger is approving too many defaulters - so having too many false negatives (FN), while not generating sufficient revenue to cover those costs.

As the threshold changes, the number of approvals and defaults change accordingly. However, without assigning value to how much profit a successful loan generates, and how much is lost per default, it is impossible to determine which threshold is optimal for profit generating.

By assigning these values, the optimal threshold can be found. First, the profit per successful loan is assumed to be £1,000, while the cost per loan that is defaulted is assumed to be £10,000. This can be used to formulate the following:

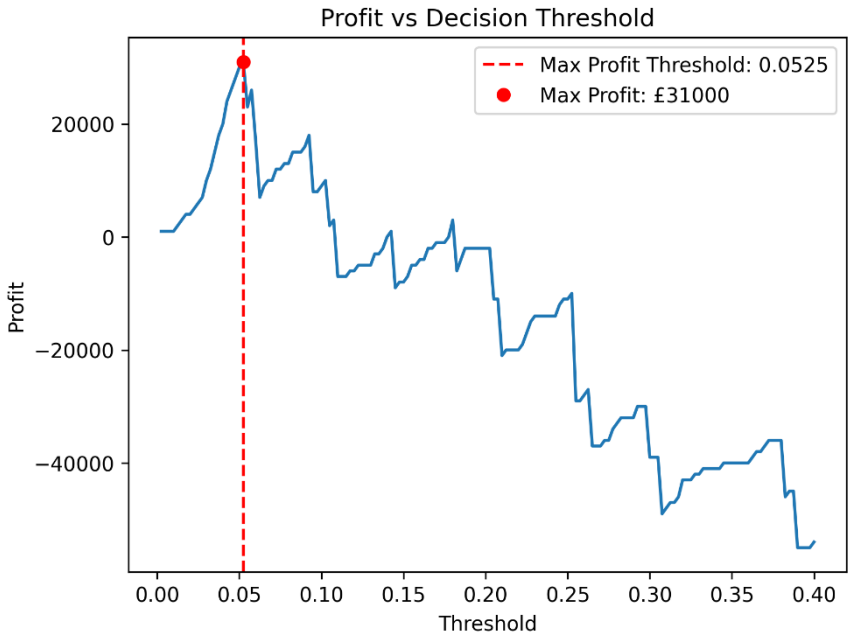
Threshold	Profit	Approvals	Approval Rate	Defaults	Default Rate
0.1	£9,000	53	26.5%	4	7.5%
0.2	£-2,000	86	43.0%	8	9.3%
0.3	£-39,000	115	57.5%	14	12.2%
0.4	£-54,000	133	66.5%	17	12.8%
0.5	£-116,000	148	74.0%	24	16.2%

Interpreting Thresholds

From the table it can be seen that the 'optimal' threshold (to 1 decimal place) is 0.1. This means rejecting a lot of good customers and being extremely cautious, only approving very low-risk borrowers.

Additionally, the importance of threshold analysis becomes apparent, as using the baseline threshold results in a significant loss.

By refining the search, more optimal thresholds can be found. The easiest way to do this is by plotting finer threshold values against the profits they would produce.



The peak in profits occurs at a threshold of approximately 0.0525, producing a profit of £31,000. This threshold is significantly lower than the initial threshold of 0.5, indicating that the model should be much more conservative when approving loans.

This reflects the difference in the profit per successful loan, against the cost of defaulting, making it optimal to reject a larger proportion of borrowers in order to minimise the risk of expensive defaults.

At this threshold, only the borrowers with very low predicted probabilities of default are approved, leading to less approvals but a higher profit. This shows how including business costs into the decision-making is crucial to effectively using the model.

Limitations

However, it is unlikely to be this simple, as loans do not all produce the same profit value, do not cost the same when defaulted, and have options outside of the binary decision of approval and rejection.

It must also be remembered that earlier, it was seen that the logistic regression model was not completely accurate and hence does not represent the true probabilities. Therefore, this threshold is unlikely to be optimal in reality, but it does make for a reasonable estimation.

It is also important to note that the profit per loan and loss per default values have been chosen arbitrarily, and changing those would inherently change the optimal threshold. Assuming a profit per loan of £1,000, and a loss per default of £5,000 the optimal threshold shifts significantly:

Threshold	Profit	Approvals	Approval Rate	Defaults	Default Rate
0.1	£29,000	53	26.5%	4	7.5%
0.2	£38,000	86	43.0%	8	9.3%
0.3	£31,000	115	57.5%	14	12.2%
0.4	£31,000	133	66.5%	17	12.8%
0.5	£4,000	148	74.0%	24	16.2%

This is consistent with expectations, as when the cost of defaulting increases, the lender should be more conservative. Therefore, when the loss is less, we should expect more leeway with the 'optimal' threshold.

Final Recommendations

Based on the analysis, logistic regression is the preferred model for this risk data. It consistently outperforms the decision tree across key evaluation metrics, including accuracy, recall for defaulters, and AUC.

While the decision tree can capture non-linear relationships, this flexibility does not result in improved predictions for this dataset. Additionally, its probability estimates are less reliable, making it less suitable for situations where decisions depend on predicted risk levels.

The threshold analysis demonstrates that the regular threshold of 0.5 is not appropriate in this context. When incorporating profits and costs, the threshold would be significantly lower, at around 0.05, reflecting the high costs when a borrower defaults.

However, this threshold depended on assumed losses and profits and would vary depending on the specific context. The model was also shown to not be perfectly calibrated, meaning the estimated probabilities may not reflect true default risk values, which would impact the threshold.

This pipeline can be improved by incorporating more realistic business assumptions, specifically variable profits and losses on loans, improved probability calibration, and exploring other modelling approaches to improve decision-making.